

Xinyuan LUO

University of Illinois at Urbana-Champaign
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EDUCATION

Autonomy and Robotics MEng,  **University of Illinois at Urbana-Champaign** Aug 2023- present

Core Courses: Introduction to Robotics(A), Principle of Safe Autonomy(A-), Human-Robot Interaction(A), Applied Machine Learning(A+), Deep Learning for robotic manipulation(A), Autonomous Vehicle System Engineering(A-). GPA: 3.85/4.0

Department of Instrument Science and Engineering,  **Shanghai Jiao Tong University** Sept 2019- Jun 2023

B.E. in Measurement Control Technology and Instruments, Overall GPA: 3.69/4.3 (12/60)

Core Courses: Robotics Foundation (96/100), Reinforcement Learning (87/100), Natural Language Processing (86/100), Image Processing and Intelligent Recognition (86/100), Principle of Automatic Control B (94/100), Principles of Sensor (95.4/100)

Honor: Endress+Hauser scholarship (3/60)

PUBLICATION

[1] X. Luo*, S. Jin*, H.-J. Huang, and W. Yuan, "An intelligent robotic system for perceptive pancake batter stirring and precise pouring," IROS 2024 Conference Paper Oral Presentation/Best Entertainment and Amusement Paper Finalist

[2] X. Luo, and W. Yuan, "A perception-manipulation robotics system for food cutting," ICRA 2024 in submission.

[3] X. Luo, and W. Yuan, "Grasping friction prediction of common groceries using tactile sensing," ICRA 2024 in submission.

RESEARCH EXPERIENCES

 **A Perception-Manipulation Robotic System for Food Cutting** [\[link\]](#)

| University of Illinois at Urbana-Champaign | Research Assistant June 2024-Sep 2024

Advisor: Wenzhen Yuan, assistant professor at the Department of Computer Science, University of Illinois at Urbana-Champaign

- Adaptive reinforcement learning controller capable of quickly generalizing to novel food and optimizing energy efficiency.
- Designing a Robotic Cutting System: knife selection, skill optimization, and performance evaluation.

 **Grasping Frictions Prediction of Common Groceries Using Tactile Sensing** [\[link\]](#)

| University of Illinois at Urbana-Champaign | Research Assistant Mar 2024-June 2024

Advisor: Wenzhen Yuan, assistant professor at the Department of Computer Science, University of Illinois at Urbana-Champaign

- Use deep learning to estimate the friction of grasping, and ensure the object is being grasped stably with minimal force.
- Fully automated data collection pipeline design.

 **An Intelligent Robotic System for Perceptive Pancake Batter Stirring and Precise Pouring** [\[link\]](#)

| University of Illinois at Urbana-Champaign | Research Assistant Sept 2023 -Mar 2024

Advisor: Wenzhen Yuan, assistant professor at the Department of Computer Science, University of Illinois at Urbana-Champaign

- Leverage Force/torque sensor to estimate viscous liquid properties including liquid level, uniformity, and viscosity.
- Achieved precise pouring based on liquid properties, with the ability to pour into custom shapes.
- Pancake-making robot stirring, perceiving, and pouring pipeline design.

 **Surgical Robot Based on Dual Position Loop Detection: Configuration Design and Motion Analysis**

| Shanghai Jiao Tong University | Research Assistant Oct 2022 -June 2023

Advisor: Hongbing Li, associate professor at Instrument Science and Engineering, Shanghai Jiao Tong University

- Design of tendon drive configuration of surgical robot.
- Research on double closed loop position detection method of tendon-drive mechanism.
- Complete the kinematic analysis of the tendon drive of the surgical robot and carry out the motion simulation analysis.

 **Human and Aerial Robot Collaborate to Carry Object**

| Shanghai Jiao Tong University | Research Assistant Oct 2021 -Aug 2022

Advisor: Hesheng Wang, professor at the Department of Automation, Shanghai Jiao Tong University

- Aerial robot assists people in moving objects compliantly while avoiding obstacles.
- Physical human-robot interaction task on the platform of aerial robot.
- Designed a variable admittance controller based on reinforcement learning. [\[link\]](#)

SKILLS

Programming Languages: Python, C/C++, Matlab, Java, HTML, CSS, JavaScript

Skills: Robot Operating System (ROS), Robot using experiences (UR5e), Pytorch, RaspberryPi, Arduino, STM32, Solidworks

Languages: Mandarin(native), English(fluent), TOEFL: Total 106 (Reading 30, Listening 27, Speaking 22, Writing 27)